glossary

alpha particle
Entity made up of two neutrons and two protons bound together by the strong nuclear force. Equivalent to a helium nucleus (a helium atom without its electrons), but such a stable unit that it acts like a particle in its own right.

asymptotic freedom
Technical term for the way the glue (or color) force between quarks decreases the closer they get together.

atom
The smallest building block of a chemical element. All atoms of any particular element are identical to each other. Each atom is made up of a tiny central nucleus, carrying positive electric charge, surrounded by a cloud of negatively charged electrons. The number of electrons in the cloud balances out the amount of positive charge in the nucleus, so the atom is electrically neutral overall.

beta radiation
Old name for a stream of fast-moving electrons.

black body
An object that is a perfect absorber of radiation.

black body radiation
Radiation emitted by a hot black body.

Bohr radius
The minimum distance at which an electron can "orbit" an atomic nucleus, in the model devised by the Dane Niels Bohr. Although the model is an imperfect description of the atom, this number—5.29×10^{-11}m—is still a good indication of the size of an atom.

cavity radiation
Old name for black body radiation.

classical mechanics
The description of the workings of the everyday world in terms of Newton's Laws and Maxwell's Equations.

classical physics
The laws of physics that apply to smoothly changing phenomena, especially Newton's Laws and Maxwell's Equations.

color
A property of quarks equivalent to the charge on electrically charged particles. Also known as color charge. It has nothing to do with color in everyday life.

Copenhagen Interpretation
The package of ideas describing the quantum world in terms of probability, uncertainty, and the "collapse of the wave function."

diffraction
The way waves bend around corners, or spread out from a small hole, as in the double-slit experiment.

electron
A fundamental entity, identified in the 1890s by the British physicist J.J. Thomson as a piece of matter that is a component of the atom. Usually thought of as a tiny particle, but one of the startling discoveries of quantum physics is that electrons also acts as waves.

element
A substance that cannot be broken down into a simpler substance by chemical means.

energy level
A quantum state with a particular energy associated with it; most commonly used to describe the quantum states available to an electron in an atom. An entity such as an electron jumps from one level to another without passing through any intermediate state; this is the famous quantum leap.

Feynman diagram
A representation of the way particles interact with each other by the exchange of force-carrying quanta, such as photons.

field
The extended influence of a force, such as electromagnetism, through space.

gamma ray
Electromagnetic radiation with very high energy, corresponding to very short wavelengths, from 10^{-10} to 10^{-14} of a meter.

gluon
Quantum entity that plays the equivalent role in quantum chromodynamics that the photon plays in quantum electrodynamics.

h
Letter used to denote Planck's constant.

half-life
The time it takes for half of the atoms (strictly speaking, nuclei) in a sample of radioactive material to decay.

interaction
Any of the four forces of nature—gravity, electromagnetism, and the strong and weak nuclear forces.

interference pattern
Pattern produced by interference, as in the double-slit experiment.

intermediate vector boson
Overall name for the three particles (W^+, W^-, and Z^0) that are the carriers of the weak interaction, carrying out the equivalent to the role of photons in the electromagnetic interaction.

jump
See quantum leap.

laser
Powerful beam of light of a single wavelength. (Light Amplification by Stimulated Emission of Radiation.)

light quantum
Photon.

matrix mechanics
A version of quantum mechanics based on equations, formulated in terms of entities known as matrices.

momentum
The force that keeps an object in motion.

neutron
One of the particles that make up the nucleus of an atom. Roughly the same mass as a proton, but with no electric charge.

Newton's laws
The laws of mechanics that apply in the everyday world, describing such things as how pool balls bounce off each other, and how rockets work.

nonlocality
The way a quantum entity, such as an electron, "spreads out" between interactions, so that it cannot be said to be localized at a point.

nucleon
Generic name for protons and neutrons, which together make up the nucleus of an atom.

nucleus
The central core of an atom, made up of positively charged protons and electrically neutral neutrons.

orbit
In general, the trajectory of any object moving under the influence of another object. In quantum physics, usually referring to the trajectory followed by an electron under the influence of the nucleus of an atom.

photoelectric effect
Ejection of electrons from a metal surface when light shines on the surface.

photon
Particle of light. Also the carrier of the electromagnetic force.

Planck's constant
Fundamental constant that relates the energy of a photon to its frequency, through the equation $E = hf$. The constant, h, appears in many equations of quantum physics.

qubit
Quantum equivalent of a bit (binary digit) in computing. A bit can only have the value 0 or 1. A qubit can also exist in a mixed state, partially 1 and partially 0.

quantum
The smallest unit of something that it is possible to have. Originally used for the quantum of light, now called a photon.

quantum chromodynamics (QCD)
Theory of how quarks interact by the exchange of gluons. Name chosen by analogy with quantum electrodynamics.